Flag Trait Outliers Within Groups

Description

Adds 'n_group_records' and 'is_trait_outlier' columns to grouped trait records using a symmetric median-IQR fence. The caller is responsible for grouping 'data_trait_records' with 'dplyr::group_by()' before calling this function. The returned data frame is always ungrouped.

Usage

```
flag_trait_outliers(

  data_trait_records,

  trait_value_column = "trait_value",

  iqr_multiplier,

  minimum_group_size = NULL

)
```

Arguments

`data_trait_records`	A grouped data frame (e.g. created by 'dplyr::group_by()'). Must contain the column named by 'trait_value_column'.
`trait_value_column`	Character scalar. Name of the numeric column to test. Default: '"trait_value"'.
`iqr_multiplier`	Positive numeric scalar. IQR fence multiplier. A value is flagged when '|value - group_median| > iqr_multiplier * group_IQR'.
`minimum_group_size`	Optional positive integer scalar. When supplied, 'is_trait_outlier' is set to 'FALSE' for groups with fewer than 'minimum_group_size' records or with an IQR of zero. Default: 'NULL' (no group-size guard applied).

Details

The IQR fence is '[group_median - iqr_multiplier * group_IQR, group_median + iqr_multiplier * group_IQR]'. Values outside this range receive 'is_trait_outlier = TRUE'.

When 'minimum_group_size' is not 'NULL', an additional guard is applied: groups with fewer than 'minimum_group_size' records or a 'group_IQR' of zero always receive 'is_trait_outlier = FALSE'.

Designed for use in pipes where the caller sets up grouping: “'r data_traits |> dplyr::group_by(trait_domain_name) |> flag_trait_outliers(iqr_multiplier = 3) “'

Value

An ungrouped data frame with the same rows and columns as 'data_trait_records' plus two additional columns:

'n_group_records'

: Integer. Number of records in the group.

'is_trait_outlier'

: Logical. ‘TRUE' when the record’s value falls outside the IQR fence; 'FALSE' otherwise. When 'minimum_group_size' is supplied, also 'FALSE' for groups that are too small or have zero IQR.

See Also

[filter_trait_outliers()], [write_trait_quality_control_report()]